

Two stochastic epidemic models

We consider two discrete-time, branching-process models for the spread of an infectious disease. Time is measured in days. Let $I_t \geq 0$ denote the number of new infections (incidence) on day t , and let $R_t > 0$ be the effective reproduction number on day t .

The *serial interval weights* $w_s > 0$ ($s = 1, 2, \dots$) satisfy $\sum_{s=1}^{\infty} w_s = 1$ and encode the probability that a secondary infection appears s days after the primary case. Write

$$F_r = \sum_{s=1}^r w_s$$

for the cumulative serial interval distribution at lag r , so $1 - F_r = \sum_{s=r+1}^{\infty} w_s$ is the probability that transmission occurs after day r .

The *dispersion parameter* $k > 0$ controls individual-level variation in infectiousness. Small k corresponds to high heterogeneity (superspreading); as $k \rightarrow \infty$ the negative-binomial offspring distribution converges to a Poisson.

Remark. Throughout, $\text{NB}(\text{mean} = \mu, \text{disp} = k)$ denotes the negative-binomial distribution with probability mass function

$$\mathbb{P}(X = n) = \binom{n+k-1}{n} \left(\frac{k}{k+\mu} \right)^k \left(\frac{\mu}{k+\mu} \right)^n, \quad n = 0, 1, 2, \dots,$$

mean μ , variance $\mu + \mu^2/k$, and probability generating function (pgf) $G(s) = (k / (k + \mu(1-s)))^k$. Similarly, $\text{Gamma}(\alpha, \beta)$ denotes the Gamma distribution with shape α and rate β , so density $f(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$, mean α/β , and variance α/β^2 .

SSE model (Superspreading Events)

In the *superspreading-events* (SSE) model, the force of infection on day t is the weighted sum of past incidence,

$$\lambda_t = \sum_{s=1}^{\infty} w_s I_{t-s}, \quad (1)$$

and new infections are negative-binomially distributed:

$$I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t). \quad (2)$$

The conditional mean of I_t is $R_t \lambda_t$ and its variance is $R_t \lambda_t + R_t^2 \lambda_t / k$, so the NB inflates the Poisson variance by the factor $1 + R_t \lambda_t / k$, capturing superspreading at the population level.

SSI model (Superspreading Individuals)

In the *superspreading individuals* (SSI) model, heterogeneity enters at the individual level. Each infected individual has a latent infectivity that is Gamma distributed with mean 1 and variance $1/k$.

Each cohort of I_t infected individuals then has an aggregate infectivity

$$Y_t \mid I_t \sim \text{Gamma}(\text{shape} = k I_t, \text{rate} = k), \quad (3)$$

so $\mathbb{E}[Y_t \mid I_t] = I_t$ and $\text{Var}(Y_t \mid I_t) = I_t / k$. New infections arise as a Poisson process driven by past infectivity,

$$I_t \mid (Y_s)_{s < t} \sim \text{Poisson} \left(R_t \sum_{s=1}^{\infty} w_s Y_{t-s} \right). \quad (4)$$

Remark. Marginalising Y_t from the SSI model recovers a negative-binomial offspring distribution (by the Poisson–Gamma mixture identity; see below), so the two models share the same mean–variance structure at the population level while differing in the *mechanism* generating overdispersion.

Extinction probability

Henceforth assume the reproduction number is constant, $R_t = R_0$. We study the probability that the epidemic goes extinct, starting from a single infectious case on day 0.

By branching-process theory, the extinction probability q is the smallest non-negative fixed point of the pgf of the offspring distribution. For both models the offspring distribution is NB(mean = R_0 , disp = k), whose pgf is $G(s) = (k/(k + R_0(1 - s)))^k$. Setting $q = G(q)$ gives

$$q = \left(1 + \frac{R_0}{k}(1 - q)\right)^{-k}. \quad (5)$$

If $R_0 \leq 1$ the unique solution is $q = 1$ (extinction is certain); if $R_0 > 1$ there is a unique solution $q \in (0, 1)$, so the probability of a major outbreak is $1 - q > 0$.

Extinction probability after observing incidence on days 0 to r

Suppose we observe incidence $I_0, I_1, \dots, I_r$ on the first $r + 1$ days, with $I_0 \geq 1$. We compute the updated extinction probability q_r for the residual epidemic — the probability that the chain triggered by the observed cases eventually dies out.

A standard branching-process argument applies in both models: each remaining offspring of the observed cases, once born, initiates an independent fresh epidemic governed by the original NB(R_0 , k) offspring distribution and goes extinct with the standard probability q (the smallest fixed point of G). The conditioning event constrains only the offspring of the observed cases on days $0, \dots, r$; it does not propagate to later generations. Therefore

$$q_r = G_r(q),$$

where G_r is the pgf of the (conditional) total remaining offspring of the observed cases. The two models differ in the form of G_r .

SSE model

In the SSE model the offspring process decomposes into independent contributions: each of the I_s individuals infected on day s generates a number of offspring at lag ℓ that is NB(mean = $R_0 w_\ell$, disp = $k w_\ell$), independently across individuals, donor days, and lags. By the NB closure property under a common success probability $p = k w_\ell / (k w_\ell + R_0 w_\ell) = k / (k + R_0)$ (independent of ℓ), the population-level NB($R_0 \lambda_t, k \lambda_t$) law of I_t is recovered by summing all per-individual-per-lag contributions.

Conditioning on the observed counts $I_0, \dots, I_r$ pins down only those contributions (s, ℓ) with $s + \ell \leq r$. Per-individual-per-lag contributions with $s \in \{0, \dots, r\}$ and $\ell > r - s$ are independent of the observations and retain their original distributions. The total remaining offspring of the observed cases is therefore

$$\sum_{s=0}^r \sum_{i=1}^{I_s} \sum_{\ell=r-s+1}^{\infty} \text{NB}(\text{mean} = R_0 w_\ell, \text{disp} = k w_\ell) = \text{NB}(\text{mean} = R_0 \Lambda, \text{disp} = k \Lambda), \quad (6)$$

where the pooled effective weight is

$$\Lambda = \sum_{s=0}^r I_s (1 - F_{r-s}). \quad (7)$$

With the convention $F_0 = 0$, the $s = r$ term enters with full weight, since none of its offspring have had a chance to appear yet.

The pgf is $G_r(s) = (1 + (R_0/k)(1 - s))^{-k\Lambda} = G(s)^\Lambda$, so

$$q_r = G_r(q) = q^\Lambda = q^{\sum_{s=0}^r I_s (1 - F_{r-s})}. \quad (8)$$

The earlier special case (a single case on day 0 followed by r days without cases) corresponds to $I_0 = 1$, $I_1 = \dots = I_r = 0$, giving $\Lambda = 1 - F_r$ and recovering $q_r = q^{1 - F_r}$. In the same direction, as the zero streak extends ($r \rightarrow \infty$) the contributions $I_s (1 - F_{r-s})$ all vanish and $q_r \rightarrow 1$: extinction approaches certainty.

SSI model: cases on day 0 only

For the SSI model a closed-form update is available only in the special case where all observed cases occur on day 0; that is, $I_0 \geq 1$ and $I_1 = \dots = I_r = 0$. We treat this case here and discuss the general situation at the end of the section.

Let y denote the realised aggregate infectivity of the I_0 index cases on day 0, with prior

$$y \sim \text{Gamma}(kI_0, k), \quad \mathbb{E}[y] = I_0. \quad (9)$$

A key observation is that $I_s = 0$ on day s forces $Y_s = 0$, since $Y_s | I_s \sim \text{Gamma}(kI_s, k)$ collapses to a point mass at 0 when $I_s = 0$. Conditioning on $I_1 = \dots = I_r = 0$ thus eliminates the day-1 through day- $(r-1)$ contributions to later infections, and so

$$I_d | y, Y_1 = \dots = Y_{d-1} = 0 \sim \text{Poisson}(R_0 w_d y), \quad d = 1, \dots, r, \quad (10)$$

with the I_d conditionally independent given y .

Step 1: posterior update given no cases on days $1, \dots, r$. By Bayes' theorem,

$$\begin{aligned} f(y | I_1 = \dots = I_r = 0) &\propto f(y) \mathbb{P}(I_1 = \dots = I_r = 0 | y) \\ &= \frac{k^{kI_0}}{\Gamma(kI_0)} y^{kI_0-1} e^{-ky} \prod_{d=1}^r e^{-R_0 w_d y} \\ &= \frac{k^{kI_0}}{\Gamma(kI_0)} y^{kI_0-1} \exp\left(-y \left[k + R_0 \sum_{d=1}^r w_d\right]\right) \\ &= \frac{k^{kI_0}}{\Gamma(kI_0)} y^{kI_0-1} e^{-(k+R_0 F_r) y}. \end{aligned} \quad (11)$$

This is proportional to a $\text{Gamma}(kI_0, k + R_0 F_r)$ density, so

$$y | (I_1 = \dots = I_r = 0) \sim \text{Gamma}(kI_0, k + R_0 F_r). \quad (12)$$

The posterior rate $k + R_0 F_r > k$ reflects the fact that observing no early transmission is evidence that y is small: the index cohort is likely a low-infectivity one.

Step 2: marginal offspring distribution for lags $s > r$. Given y , the total remaining offspring is

$$I_{\text{rem}} | y = \sum_{s=r+1}^{\infty} I_s \sim \text{Poisson}(R_0(1 - F_r) y). \quad (13)$$

Setting $\lambda = R_0(1 - F_r) y$, the Gamma scaling property ($cX \sim \text{Gamma}(\alpha, \beta/c)$ for $X \sim \text{Gamma}(\alpha, \beta)$) gives

$$\lambda \sim \text{Gamma}\left(kI_0, \frac{k + R_0 F_r}{R_0(1 - F_r)}\right), \quad (14)$$

and the Poisson–Gamma mixture identity yields

$$I_{\text{rem}} \sim \text{NB}\left(\text{size} = kI_0, \text{mean} = kI_0 \cdot \frac{R_0(1 - F_r)}{k + R_0 F_r}\right). \quad (15)$$

The effective per-individual mean $R_0(1 - F_r)/(1 + R_0 F_r/k)$ is reduced relative to the unconditional $R_0(1 - F_r)$ by the same Bayesian-update factor as in the $I_0 = 1$ case.

Step 3: extinction probability. Each remaining offspring, once born, initiates an independent fresh epidemic with infectivity drawn from the prior $\text{Gamma}(k, k)$, so it is governed by the original $\text{NB}(R_0, k)$ offspring distribution and goes extinct with the standard probability q . Hence

$$q_r = G_r(q) = \left(1 + \frac{R_0(1 - F_r)}{k + R_0 F_r}(1 - q)\right)^{-kI_0}. \quad (16)$$

For $I_0 = 1$ this reduces to the previous formula.

SSI model: general histories (MCMC)

When at least one $I_d > 0$ for $d \in \{1, \dots, r\}$, the day- d rate involves $R_0 \sum_{s=1}^{d-1} w_{d-s} Y_s$ — a sum of latent infectivities — so the joint posterior of $(Y_0, \dots, Y_{r-1})$ is no longer a product of independent Gammas and the marginal offspring distribution does not admit a closed form. We therefore estimate q_r by MCMC.

MCMC posterior. We sample $(Y_0, \dots, Y_{r-1})$ from their joint posterior given $(I_0, \dots, I_r)$ using a Hamiltonian Monte Carlo sampler (implemented in PyMC). Note that Y_r , the aggregate infectivity of the day- r cohort, has not yet contributed to any observed outcome; its posterior equals its prior,

$$Y_r \sim \text{Gamma}(kI_r, k),$$

and is sampled independently.

Remaining offspring from observed cases. Given $(Y_s)_{s=0}^r$, the future infections generated by the observed cases arrive as independent Poisson streams: cohort s contributes at lag $\ell \geq r - s + 1$ at rate $R_0 w_\ell Y_s$. Summing over cohorts and lags gives total remaining offspring

$$I_{\text{rem}} \mid (Y_s)_{s=0}^r \sim \text{Poisson}(R_0 \Lambda), \quad \Lambda = \sum_{s=0}^r Y_s (1 - F_{r-s}), \quad (17)$$

exactly the same pooled-weight formula as in the SSE model.

Extinction probability and MCMC estimator. Each remaining offspring initiates an independent fresh epidemic that goes extinct with the standard probability q . Hence the conditional extinction probability of the residual epidemic, given the latent infectivities, is the pgf of $\text{Poisson}(R_0 \Lambda)$ evaluated at q :

$$q_r \mid (Y_s)_{s=0}^r = e^{R_0 \Lambda (q-1)}. \quad (18)$$

Marginalising over the posterior of (Y_s) and estimating by a Monte Carlo average over M MCMC draws,

$$\hat{q}_r = \frac{1}{M} \sum_{m=1}^M e^{R_0 \Lambda^{(m)} (q-1)}, \quad \widehat{\text{PMO}} = 1 - \hat{q}_r, \quad (19)$$

where $\Lambda^{(m)} = \sum_{s=0}^r Y_s^{(m)} (1 - F_{r-s})$.

Remark (Consistency with the day-0-only analytic formula). When $I_1 = \dots = I_r = 0$ the observations force $Y_1 = \dots = Y_r = 0$, so $\Lambda = Y_0(1 - F_r)$ and the posterior is $Y_0 \sim \text{Gamma}(kI_0, k + R_0 F_r)$. Taking the expectation of $e^{R_0 Y_0 (1 - F_r)(q-1)}$ with respect to this posterior via the Gamma moment-generating function gives

$$q_r = \left(1 - \frac{R_0(1 - F_r)(q-1)}{k + R_0 F_r}\right)^{-kI_0} = \left(1 + \frac{R_0(1 - F_r)}{k + R_0 F_r}(1 - q)\right)^{-kI_0},$$

which matches the closed-form result in equation (11).

Probability of major outbreak under model uncertainty

So far we have computed the probability of a major outbreak conditional on a chosen model (SSE or SSI). In practice the underlying mechanism generating overdispersion is unknown, and we may wish to combine the two via Bayesian model averaging. Let $\pi_{\text{SSE}}, \pi_{\text{SSI}} \in [0, 1]$ with $\pi_{\text{SSE}} + \pi_{\text{SSI}} = 1$ be prior probabilities on the two models. Given the observed history $(I_0, \dots, I_r)$, Bayes' theorem yields posterior model probabilities

$$\pi_M^* = \frac{\pi_M L_M}{\pi_{\text{SSE}} L_{\text{SSE}} + \pi_{\text{SSI}} L_{\text{SSI}}}, \quad L_M = \mathbb{P}_M(I_1, \dots, I_r \mid I_0), \quad M \in \{\text{SSE}, \text{SSI}\}. \quad (20)$$

The model-averaged probability of a major outbreak is then

$$\text{PMO}_{\text{avg}} = \pi_{\text{SSE}}^* \cdot \text{PMO}_{\text{SSE}} + \pi_{\text{SSI}}^* \cdot \text{PMO}_{\text{SSI}}, \quad (21)$$

where $\text{PMO}_M = 1 - q_r^M$ as derived above.

Likelihood of the observed history

SSE model. Conditional on past incidence, the day- t count satisfies $I_t \mid I_{<t} \sim \text{NB}(\text{mean} = R_0 \lambda_t, \text{disp} = k \lambda_t)$ with $\lambda_t = \sum_{s=1}^{t-1} w_{t-s} I_s$, so the joint likelihood factorises into a product of negative-binomial pmfs,

$$L_{\text{SSE}}(I_1, \dots, I_r \mid I_0) = \prod_{t=1}^r \text{NB}(I_t; R_0 \lambda_t, k \lambda_t).$$

This closed form is straightforward to compute for any history; we restrict attention to the day-0-only special case below for the analytic implementation.

SSI model. The likelihood involves the latent infectivities $(Y_s)_{s=0}^{r-1}$ and is in general not available in closed form. For the special case $I_1 = \dots = I_r = 0$ a closed form follows directly from the Gamma moment-generating function (below); for general histories an importance-sampling estimate using draws from the posterior $p(Y_{0:r-1} \mid I_{0:r})$ produced by the MCMC sampler of the previous section is feasible but is not implemented here.

Closed form for cases on day 0 only

Take $I_0 \geq 1$ and $I_1 = \dots = I_r = 0$.

For the SSE model, $I_t \sim \text{NB}(R_0 I_0 w_t, k I_0 w_t)$ independently across $t = 1, \dots, r$. The probability that all are zero factorises as

$$L_{\text{SSE}} = \prod_{t=1}^r \left(\frac{k I_0 w_t}{k I_0 w_t + R_0 I_0 w_t} \right)^{k I_0 w_t} = \left(\frac{k}{k + R_0} \right)^{k I_0 F_r} = \left(1 + \frac{R_0}{k} \right)^{-k I_0 F_r}. \quad (22)$$

For the SSI model, conditional on $Y_0 \sim \text{Gamma}(k I_0, k)$ the day- t count is independently Poisson($R_0 w_t Y_0$) for $t = 1, \dots, r$, so $\mathbb{P}(I_1 = \dots = I_r = 0 \mid Y_0) = \exp(-R_0 F_r Y_0)$. Marginalising via the Gamma MGF gives

$$L_{\text{SSI}} = \mathbb{E}[\exp(-R_0 F_r Y_0)] = \left(1 + \frac{R_0 F_r}{k} \right)^{-k I_0}. \quad (23)$$

Both likelihoods share the same functional shape but with different exponents: the SSE exponent $k I_0 F_r$ is reduced by the factor $F_r \leq 1$ relative to the SSI exponent $k I_0$, reflecting the fact that the SSE total over days $1, \dots, r$ has dispersion $k I_0 F_r$ whereas the SSI total has dispersion $k I_0$. Combining with the per-model PMOs from earlier sections gives

$$\text{PMO}_{\text{avg}} = \pi_{\text{SSE}}^* (1 - q^{I_0(1-F_r)}) + \pi_{\text{SSI}}^* \left[1 - \left(1 + \frac{R_0(1-F_r)}{k + R_0 F_r} (1 - q) \right)^{-k I_0} \right]. \quad (24)$$

Remark. When $r = 0$ (no observations after day 0) we have $F_r = 0$ and $L_{\text{SSE}} = L_{\text{SSI}} = 1$, so the posterior reduces to the prior and the model-averaged PMO is just the prior-weighted average of the two per-model PMOs.

Simulation-based estimator

For general histories, where the SSI likelihood is unavailable in closed form, we use a rejection-sampling estimator that simultaneously realises the posterior model weighting. Repeat the following in batches of size b :

1. Draw the per-batch model split as a Binomial: $b_{\text{SSE}} \sim \text{Bin}(b, \pi_{\text{SSE}})$ and $b_{\text{SSI}} = b - b_{\text{SSE}}$.
2. Run b_{SSE} vectorised SSE forward simulations from $t = 0$ seeded with I_0 index cases, retaining only those whose simulated incidence on days $1, \dots, r$ exactly matches the observed $I_1, \dots, I_r$. Likewise for b_{SSI} SSI simulations.

3. For each retained simulation, continue forward until either the single-step incidence reaches the major-outbreak threshold (resolved as “major”) or the most recent $L = \text{len}(w)$ steps are all zero (resolved as “extinct”).

Continue accumulating batches until a target number of resolved-and-matching simulations is reached. The acceptance rate of model M ’s simulations is L_M in expectation; with a Binomial split by prior, the expected number of accepted M -sims is proportional to $\pi_M L_M$, so the composition of the accepted pool matches the posterior π_M^* . The pooled estimator

$$\widehat{\text{PMO}}_{\text{avg}} = \frac{N_{\text{SSE}}^{\text{maj}} + N_{\text{SSI}}^{\text{maj}}}{N_{\text{SSE}}^{\text{acc}} + N_{\text{SSI}}^{\text{acc}}} = \hat{\pi}_{\text{SSE}}^* \widehat{\text{PMO}}_{\text{SSE}} + \hat{\pi}_{\text{SSI}}^* \widehat{\text{PMO}}_{\text{SSI}} \quad (25)$$

follows immediately, where $\hat{\pi}_M^* = N_M^{\text{acc}} / (N_{\text{SSE}}^{\text{acc}} + N_{\text{SSI}}^{\text{acc}})$ is the implied posterior estimate and $\widehat{\text{PMO}}_M = N_M^{\text{maj}} / N_M^{\text{acc}}$ is the per-model PMO from the accepted M -sims.